

## Practical – Week2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
sanjay_dundra@Sanjay:~$ nano filecopy.c
sanjay_dundra@Sanjay:~$ echo "This is the content of the source file." > source.txt
sanjay_dundra@Sanjay:~$ cat source.txt
This is the content of the source file.
sanjay_dundra@Sanjay:~$ gcc filecopy.c -o filecopy
sanjay_dundra@Sanjay:~$ ./filecopy
File copied successfully.
sanjay_dundra@Sanjay:~$ cat destination.txt
This is the content of the source file.
sanjay_dundra@Sanjay:~$ ls -l source.txt destination.txt
-rw-r--r-- 1 sanjay_dundra sanjay_dundra 40 Aug 14 10:05 destination.txt
-rw-r--r-- 1 sanjay_dundra sanjay_dundra 40 Aug 14 10:05 source.txt
sanjay_dundra@Sanjay:~$ diff source.txt destination.txt
sanjay_dundra@Sanjay:~$ _
```

Report: I compiled and ran `filecopy.c` to copy `source.txt` to `destination.txt`. The program reported "File copied successfully," and `cat destination.txt` showed the content matched the source exactly. `ls -l` confirmed both files were the same size (40 bytes), and `diff source.txt destination.txt` produced no output, indicating no differences. This confirms the program correctly reads from the source file and writes an identical copy to the destination file.

2. Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

```

sanjay_dundra@Sanjay:~$ cat sample.txt
Hello Linux System Calls
sanjay_dundra@Sanjay:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc8b9a6738 /* 26 vars */)
= 0
brk(NULL)                                = 0x628c2b6c4c000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ed7fe2344000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20943, ...}) = 0
mmap(NULL, 20943, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7ed7fe22e000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ed7fe1fb000
mmap(0x7ed7fe202000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7ed7fe202000
mmap(0x7ed7fe223000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7ed7fe223000
mmap(0x7ed7fe22a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7ed7fe22a000
mmap(0x7ed7fe22c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7ed7fe22c000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ed7fe1cd000
mmap(0x7ed7fe1d1000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7ed7fe1d1000

```

```

read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ed7fde45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ed7fde45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ed7fde45cb0}, {sa_handler=0x62c28287f0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ed7fde45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ed7fde45cb0}, {sa_handler=0x62c28287f0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ed7fde45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x62c2b6c8e000) = 0x62c2b6c8e000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=25, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 25
splice(4, NULL, 1, NULL, 25, 0Hello Linux System Calls) = 25
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7ed7fe231000, 12288) = 0

```

Report: I ran `strace cat sample.txt`. The trace showed `openat()` to open the file, `fstat()` for metadata, `splice()` to transfer data directly to stdout, and `close()` to release the descriptor. This confirms `cat` uses efficient kernel calls like `splice()` rather than a plain read/write loop.